

ENGUANG FAN

enguang2@illinois.edu · enguang2.github.io · Google Scholar

EDUCATION

University of Illinois, Urbana-Champaign, IL, USA

Aug 2024 – May 2028 (expected)

Ph.D. Candidate in Computer Science

Advisor: Prof. Klara Nahrstedt

Research: Spatial Intelligence, Wireless and Mobile Systems, Machine Learning, and Localization

University of Illinois, Urbana-Champaign, IL, USA

Aug 2022 – May 2024

Master of Computer Science

University of Illinois, Urbana-Champaign, IL, USA

Aug 2019 – Aug 2022

*B.S. in Statistics & Computer Science, **Highest Distinction***

EXPERIENCE

Student Researcher (Intern), Google

May 2026 – Aug 2026, Mountain View, CA

Android Location Team — the team behind indoor positioning in Google Maps and Android Location Services.

Supervised by Dr. Roy Want

- Advanced the integration of Wi-Fi Round Trip Time (RTT) indoor localization into Android's *Fused Location Provider (FLP)* and scaled evaluation across numerous Google offices in Mountain View.
- Achieved **sub-second localization latency** with **1 m 90th-percentile error** on Google Wi-Fi access points and **2 m** on enterprise-grade HPE Aruba, Cisco, and Arista wireless access points.
- Developed and evaluated tightly coupled particle-filter and Bayesian-grid methods for fusing Wi-Fi RTT measurements with motion sensors across varied motion and ranging conditions.
- Designed and implemented *APSelector*, a geometry-aware AP selection algorithm that prioritizes access points providing strong geometric constraints to mitigate NLOS and positive ranging biases.
- Incorporated IMDF walkable-path constraints, when available, as probabilistic priors to reject infeasible trajectories and further improve localization accuracy.

Student Researcher (Intern), Google

May 2025 – Aug 2025, Mountain View, CA

Android Location Team — the team behind indoor positioning in Google Maps and Android Location Services.

Supervised by Dr. Roy Want

- Designed and developed an advanced indoor localization prototype combining Wi-Fi Round Trip Time (RTT) and motion sensor fusion on Pixel phones, achieving substantially lower latency and higher accuracy than Google's production *Fused Location Provider (FLP)* under challenging indoor environments (DFS channels, multipath-heavy areas).
- Implemented a passive Wi-Fi listening framework that reuses cached scan results from FLP to enable continuous RTT-based ranging without active scans, reducing localization update **latency by 57%** and eliminating blackout periods.
- Built a high-precision pedestrian dead reckoning (PDR) module leveraging Android's step detector and orientation sensors, and fused it with Wi-Fi RTT measurements through a multi-state Kalman filter, achieving **0.8 meter indoor localization accuracy** at 90% CDF using existing commercial Wi-Fi infrastructure.
- Contributed to ongoing efforts to integrate Wi-Fi RTT capabilities into Google's Fused Location Provider framework for next-generation indoor localization.

PUBLICATIONS

- [1] **Enguang Fan** et al. "RTT-MOTION: Scan-Aware Real-Time Indoor Localization with Commodity Wi-Fi RTT." Under submission to *ACM MobiCom 2027*.
- [2] **Enguang Fan** et al. "Learning Latent RF Landmarks from Unlabeled Mobile Trajectories." Under submission to *NeurIPS 2026*.

- [3] **Enguang Fan**, Yifan Chen, Zihan Shan, Matthew Caesar, Jae Kim. “*Communication-Aware Multi-Agent Reinforcement Learning for Decentralized Cooperative UAV Deployment*.” Accepted to *IEEE GLOBECOM 2026*.
- [4] **Enguang Fan**, Anfeng Peng, Matthew Caesar, Jae H Kim, Josh Eckhardt, Greg Kimberly, Denis Osipychiev. “*Towards Effective Swarm-Based GPS Spoofing Detection in Disadvantaged Platforms*.” In *IEEE MILCOM 2023*.
- [5] Emerson Sie*, **Enguang Fan***, Federico Cifuentes-Urtubey, Deepak Vasisht. “*Crowdsourcing Ubiquitous Indoor Localization with Non-Cooperative Wi-Fi Ranging*.” *arXiv preprint*, 2025.
- [6] **Enguang Fan***, Emerson Sie*, Federico Cifuentes-Urtubey, Deepak Vasisht. “*Scalable Indoor Localization with Non-Cooperative Wi-Fi Ranging*.” Poster at *ACM MobiCom 2025*. **Best Poster Runner-Up**.
- [7] **Enguang Fan***, Binh Minh Tran*, Klara Nahrstedt. “*Meter-Level Wi-Fi RTT Localization on a Production Enterprise WLAN*.” In *ACM MobiCom 2026* (short paper).
- [8] **Enguang Fan**. “*Real-Time Loop Closure Detection in Visual SLAM via NetVLAD and Faiss*.” *arXiv preprint*, 2026.

RESEARCH EXPERIENCE

Learning Latent RF Landmarks from Unlabeled Mobile Trajectories

Feb 2026 – Present

- Developed a self-supervised Radio Frequency (RF) place-recognition framework that learns discriminative spatial embeddings from sparse Wi-Fi RTT/RSS trajectory segments, enabling mobile devices to discover repeatable latent landmarks without labeled locations or known access-point coordinates.
- Introduced RF-based loop closure for crowdsourced mobile SLAM, matching independently collected trajectory segments and incorporating high-confidence RF correspondences as constraints in pose-graph optimization to correct accumulated pedestrian dead-reckoning drift.
- Designed an end-to-end mapping pipeline that jointly reconstructs mobile trajectories and reusable RF landmarks from unlabeled walks, with robustness to missing AP observations, multipath, trajectory direction, and cross-device measurement variation.
- Built and released a synchronized Wi-Fi RTT/RSS/IMU trajectory dataset and reproducible SLAM benchmark with ground-truth poses for evaluating RF place recognition, loop-closure detection, and crowdsourced spatial mapping.

Crowdsourcing Ubiquitous Indoor Localization with Wi-Fi Ranging

Oct 2024 – May 2025

Supervised by Prof. Deepak Vasisht

- Designed and implemented *PeepLoc*, a wireless indoor localization system fusing non-cooperative Wi-Fi ranging with pedestrian dead reckoning (PDR) without infrastructure support or PHY-layer access, achieving **56% lower mean error** than Android’s Fused Location Provider (3.41 m vs. 7.71 m) across four buildings.
- Proposed a probabilistic backend that geolocates access points by fusing one-way time-of-flight (ToF) estimates with PDR trajectories in a joint non-linear least squares problem under clock-offset uncertainty.
- Developed a per-AP adaptive ranging model correcting non-line-of-sight (NLOS) multipath distortion and hardware-induced RTT slope deviations, improving end-to-end localization accuracy indoors.

HONORS

- MobiCom 2025 Best Poster Runner-Up
- UIUC Fall 2022 Teachers (TA) Ranked as Excellent by Their Students [Link]

SKILLS

- Languages: Python, C, C++, Java, MATLAB
- Machine Learning, Deep Learning & Computer Vision: PyTorch, TensorFlow, Transformers, LSTMs, Reinforcement Learning, OpenCV
- Robotics & State Estimation: SLAM, visual-inertial odometry, factor graphs, Kalman/particle filters, sensor fusion, PDR, ROS/ROS2
- Wireless & Mobile Systems: Wi-Fi RTT, Android, Linux, CMake